



PORTFOLIO

NASTASSIA SYSOYEVA

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Bird's eye view on the site

WOHNEN AM STEIGER

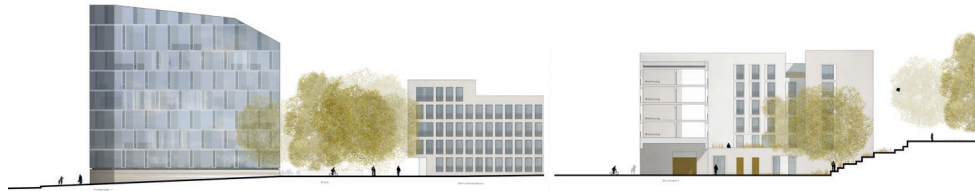
REALIZATION COMPETITION
HOFFMANN SEIFERT PARTNER ARCHITEKTEN INGENIERE

The project forms the urban conclusion at the southern edge of Erfurt, marking the transition to the Steigerwald forest, and responds sensitively to noise, topography, and existing structures. Along Arnstädter Str. and Martin-Andersen-Nexö-Str., clearly defined, noise-shielding edges are created, while the district opens to the west and south for future developments. A freestanding office tower serves as a distinctive landmark at the city's entrance. Despite its introverted overall character, the neighborhood connects with adjacent urban areas through appropriate scale, clear addressing, and differentiated open spaces.

Within the site, freestanding buildings and urban villas structure the district into manageable neighborhoods with courtyard-like open spaces. The var-

ied topography is strategically utilized to minimize construction effort, efficiently integrate underground parking, and create a well-defined central area. Access is provided via an extension of Grimmstraße and internal residential streets, complemented by pedestrian and bicycle paths linking to the urban forest.

The program combines townhouses, apartment buildings with commercial uses, senior housing, and office space, offering high flexibility and strong market potential. Sustainability is ensured through district heating, reduced surface sealing, green roofs, and a differentiated rainwater management system. An integrated strategy for noise mitigation, species protection, and drainage safeguards the long-term urban and ecological quality of the neighborhood.

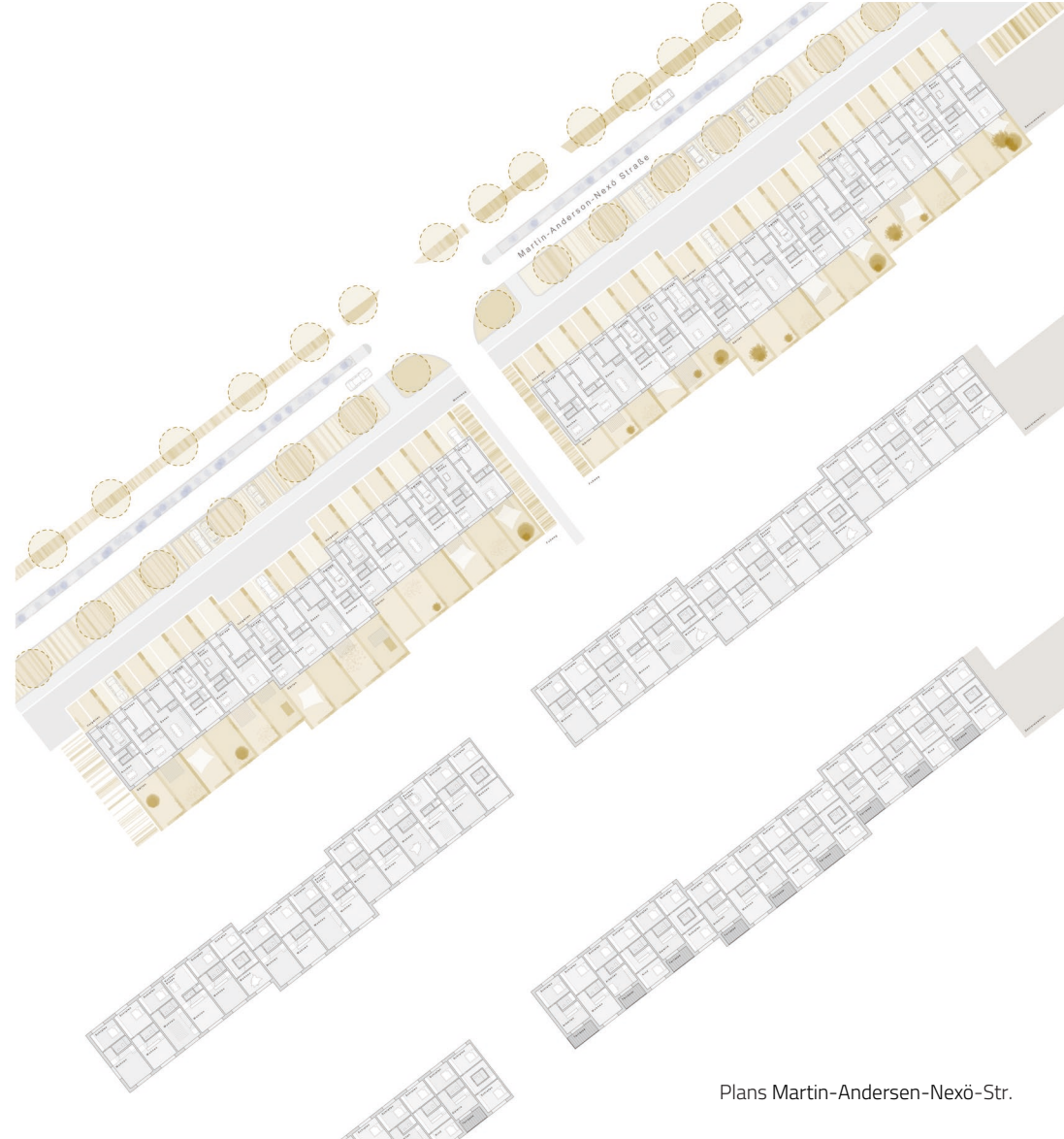
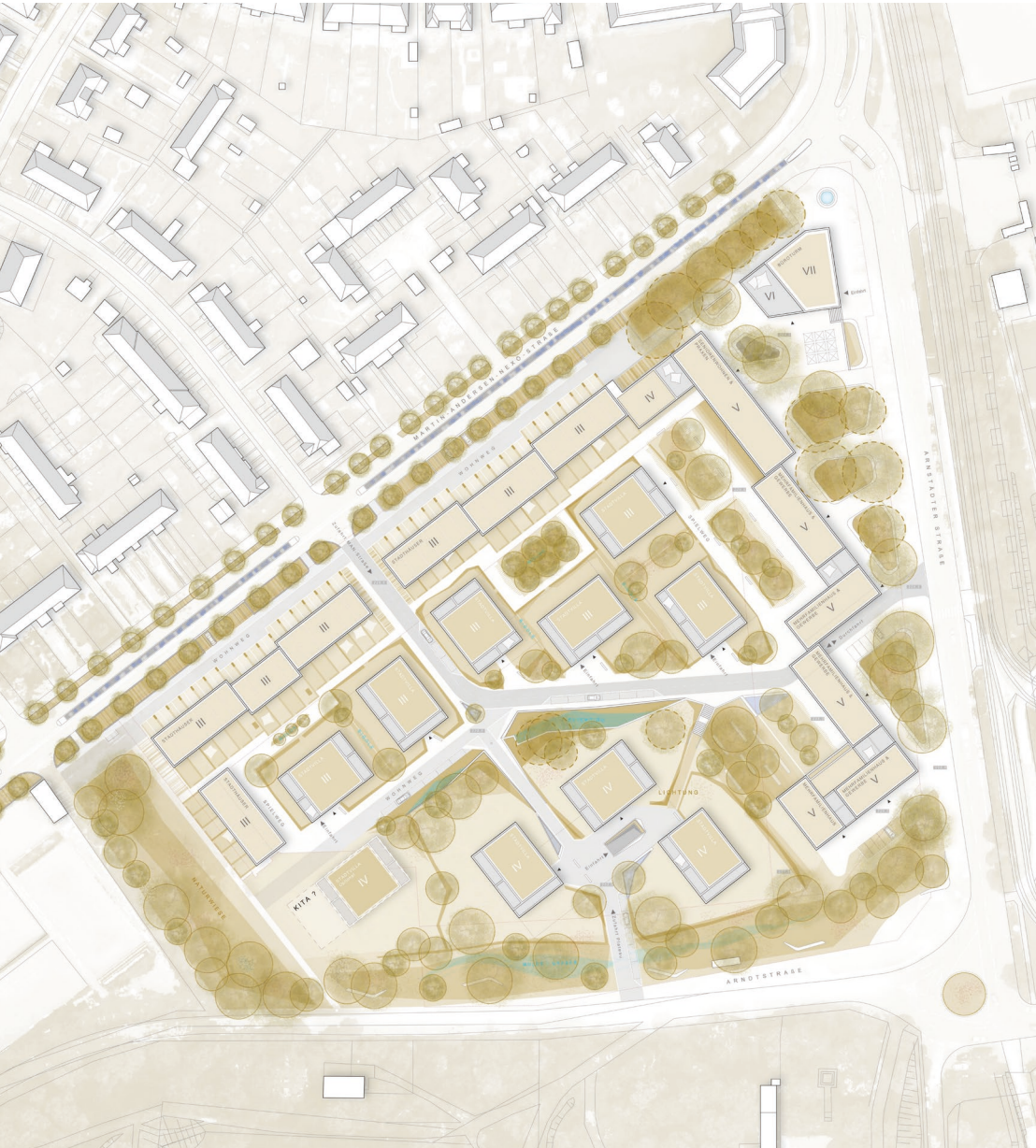


View office tower north-west

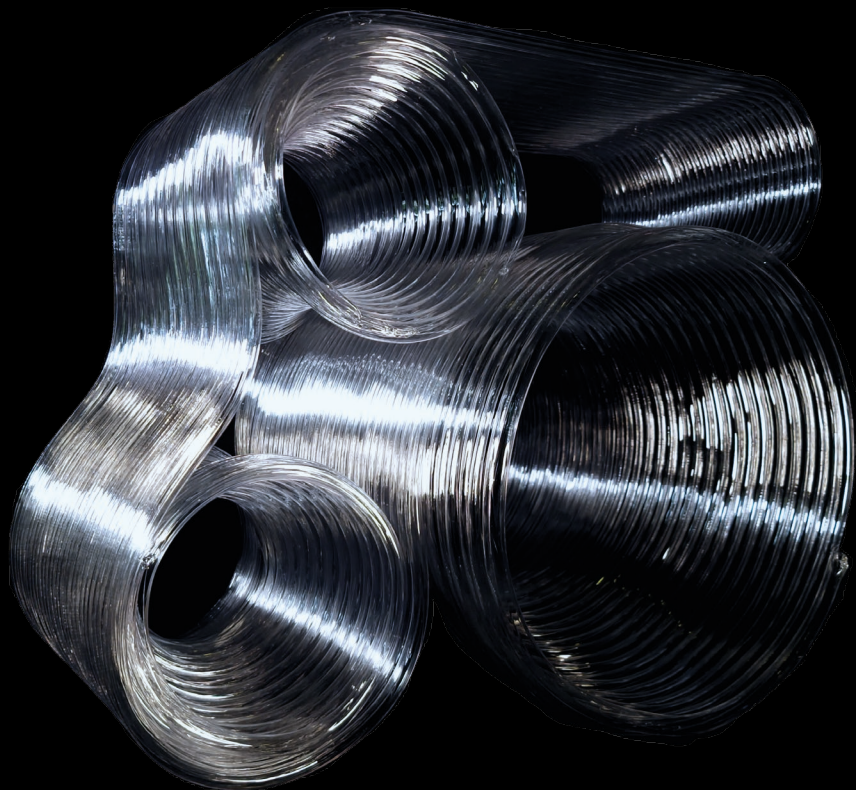
Section view Arnstädter Str. north-west



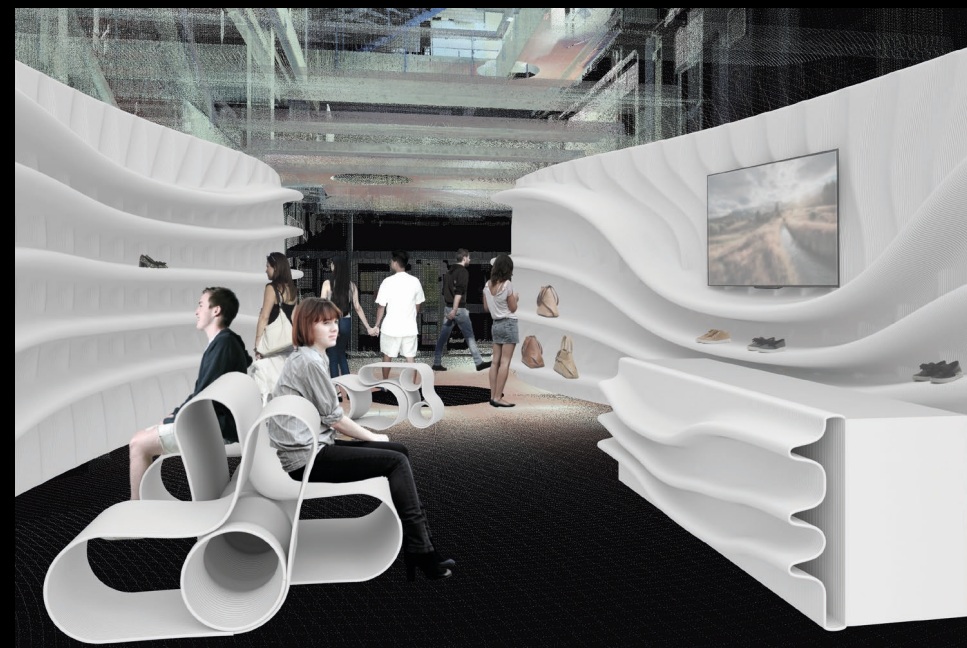
View Martin-Andersen-Nexö-Str. north-west



Plans Martin-Andersen-Nexö-Str.



Printed chair-prototype with a side-table



HOLLOW TECTONICS

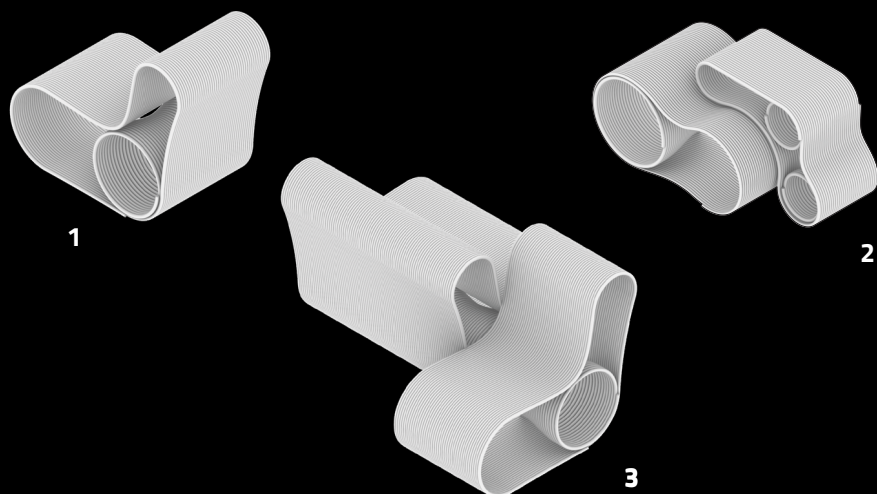
MASTER-THESIS
SUPERVISOR PROF. OLIVER TESSMANN
SUMMER SEMESTER 2025
TECHNICAL UNIVERSITY OF DARMSTADT, DIGITAL DESIGN UNIT (DDU)

The conventional FDM printing process with polymers produces solid strands, causing models and prototypes to gain significant weight as wall thickness increases. In large-scale 3D printing for architecture and furniture design, this results in high material consumption, rising costs, and increased plastic waste and CO₂ emissions.

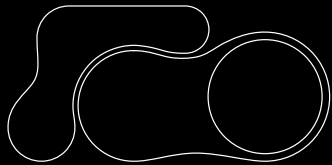
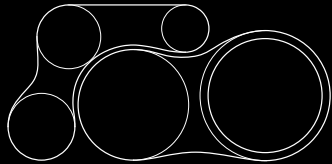
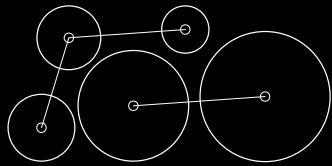
The FFBX method enables the 3D printing of architectural designs with reduced material use, lower component weight, and shorter printing times, even at large layer heights. Using blow extrusion and robotic fabrication, I am investigating this process in the field of interior design, where plastic printing is

already applied to partitions, screens, and retail interiors.

The aim of my master's thesis is to demonstrate blow extrusion as an alternative fabrication method with significantly reduced weight and material consumption, while also exploring its aesthetic and functional potential. This includes designs with large layer heights (8–20 mm), hollow and fillable strands, varying print paths for ornamentation, and variable cross-sections. These aspects are being tested through the design and 3D printing of objects at a scale between furniture and interior architecture, such as partition walls.

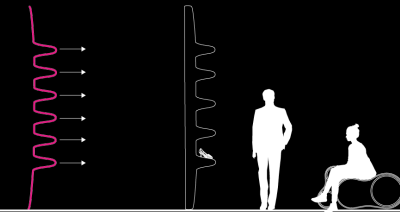
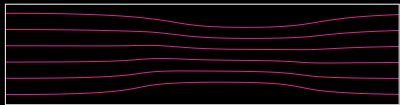
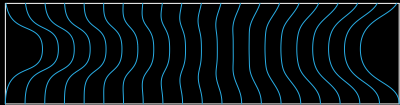


Possible configurations: 1. chair; 2. chair with a side-table; 3. stacked chairs

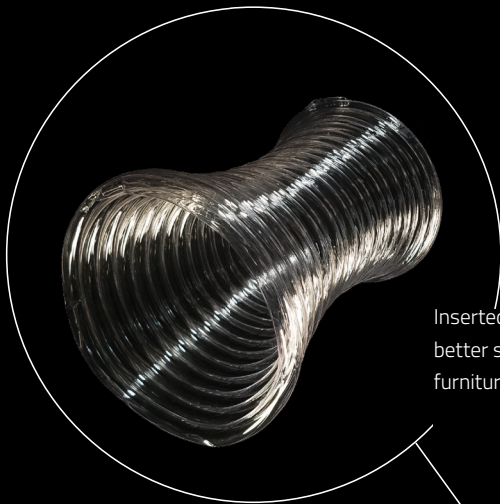


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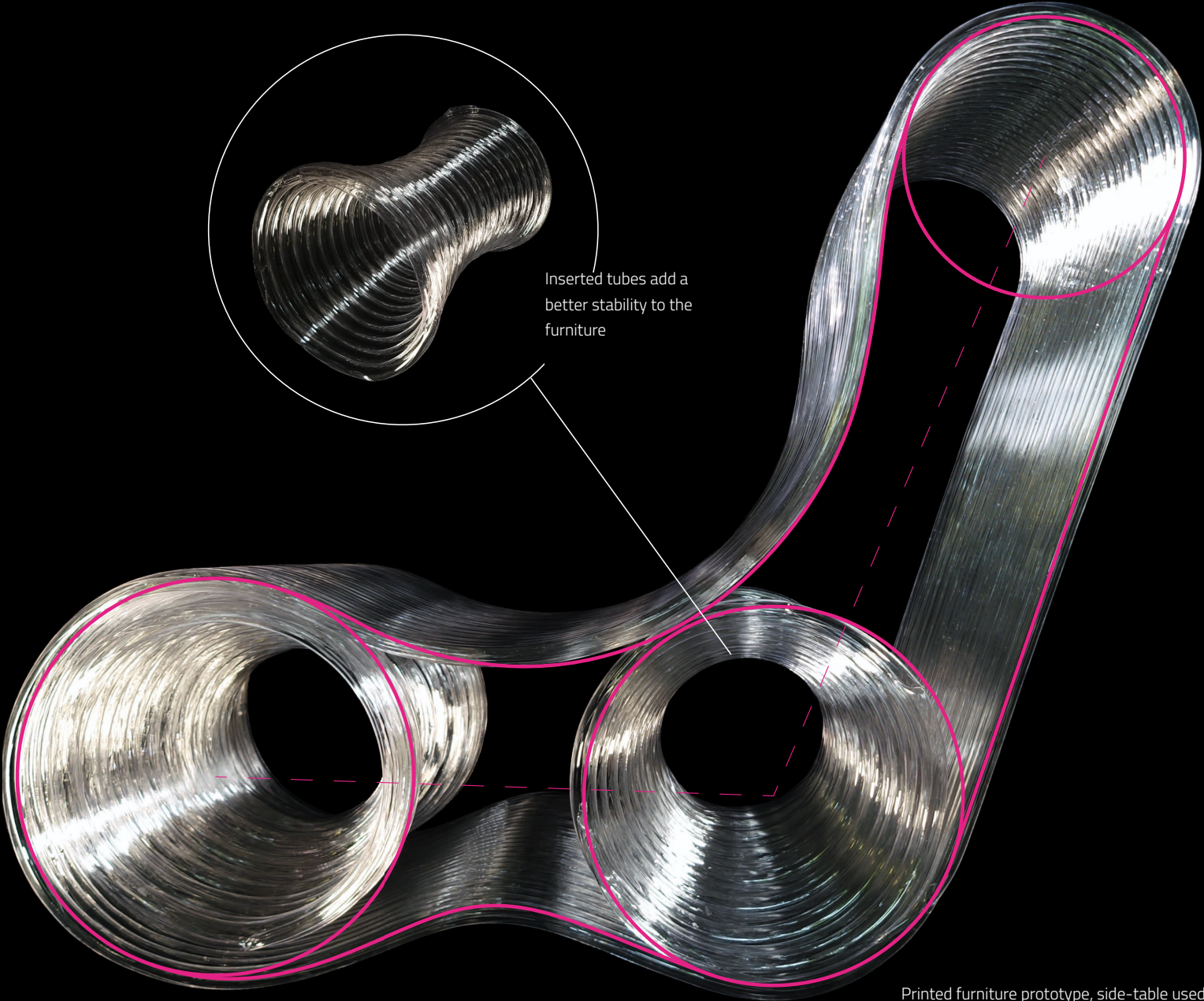
Furniture geometry



Wall geometry

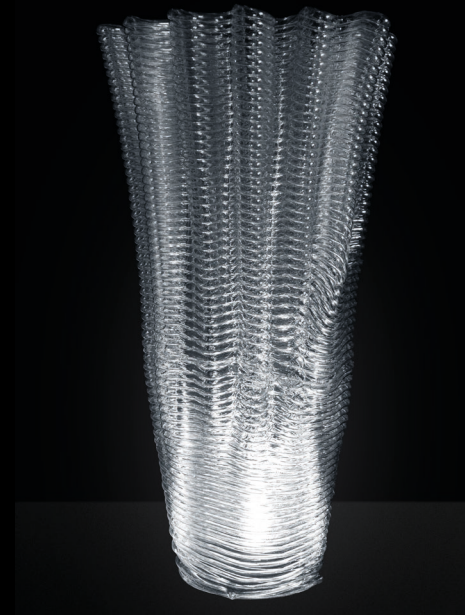


Inserted tubes add a better stability to the furniture

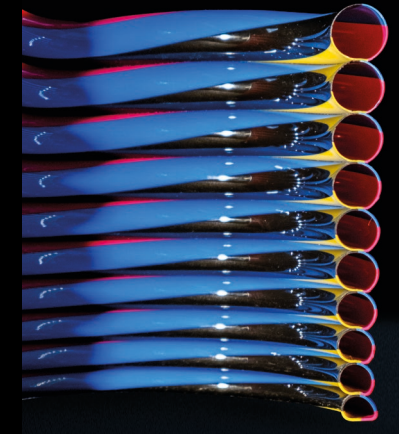


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Printed furniture prototype, side-table used as a chair



First column prototype



FFF-print with a variable layer height and multicolor filaments

MULTI-MATERIAL HOLLOW-STRANDS BY BLOW EXTRUSION (BX) 3D PRINTING

SAMIM MEHDIZADEH, PHILIPP WÜST, NASTASSIA SYSOYEVA, DIETER SPIEHL,
PROF. ANDREAS BLAESER, PROF. OLIVER TESSMANN

Blow-Extrusion (BX) 3D printing is an innovative additive manufacturing method that enables the production of three-dimensional structures from hollow strands. The process combines Fused Granular Fabrication (FGF) and Fused Filament Fabrication (FFF) with multi-material 3D printing, robotic fabrication, and computational design. By using variably adjustable cross-sections of the hollow strands, material and resource consumption can be significantly reduced.

Particularly in large-scale 3D fabrication with thermoplastics for applications in the architecture, engineering, and construction (AEC) industry, BX 3D printing offers substantial advantages. Compared to

fully solid cross-sections, material savings of up to 85% can be achieved while still producing lightweight yet load-bearing components. The precise control of airflow during extrusion further enables high adaptability and efficiency in large-scale manufacturing.



Wall elements

ADDITIVE HYBRIDS

TECHNICAL UNIVERSITY OF DARMSTADT, DIGITAL DESIGN UNIT (DDU)

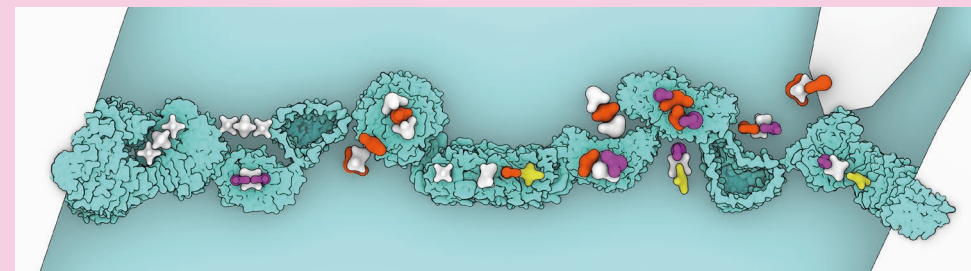
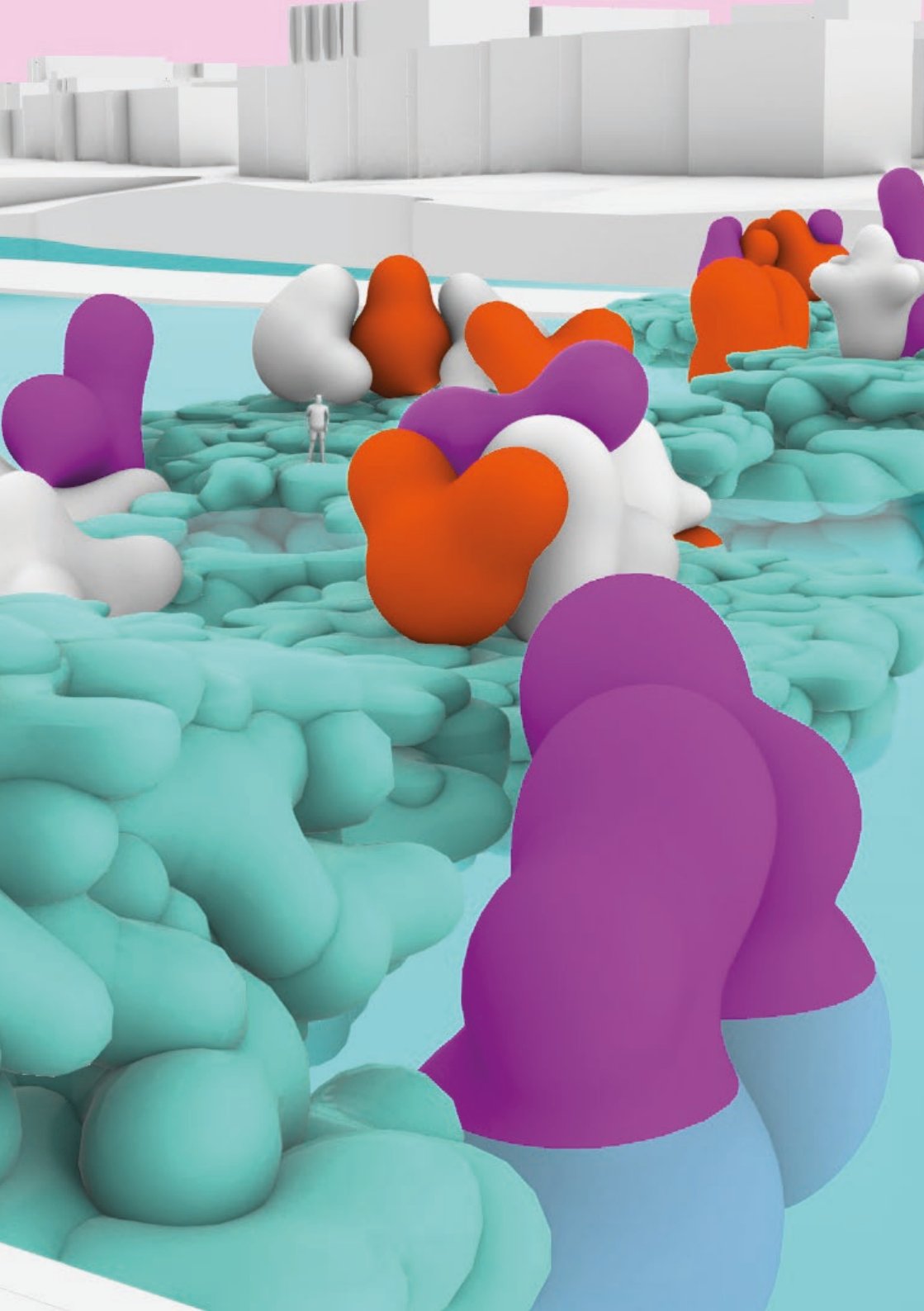
SAMIM MEHDIZADEH, NASTASSIA SYSOYEVA, MOLHAM JORBAH, LEON WITSCHORKE, PHILIPP WÜST, OLIVER TESSMANN

"Additive Hybrids" is a demonstrator that investigates the potential of additive manufacturing in combination with various digital fabrication techniques and multi-material approaches. The use of 3D printing with building materials at the scale of construction components opens up new possibilities for hybrid materiality. Research in large-scale additive manufacturing continues to address key challenges, particularly the integration of multiple materials.

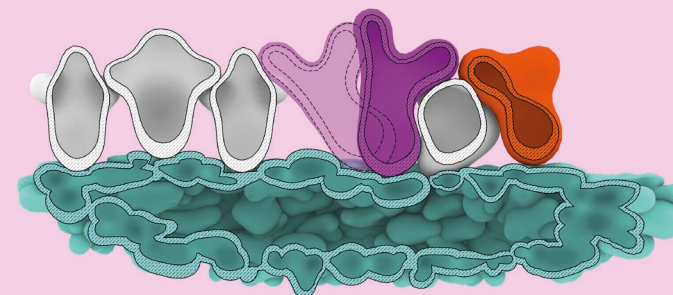
Against this backdrop, we present a hybrid material system that simultaneously combines 3D printing and casting processes. The system employs a bio-based plastic (PET), characterized by high recyclability, together with a water-based acrylic polymer as

the casting material. Our algorithmic design framework optimizes material efficiency by strategically placing each material only where it is needed, thereby minimizing material overlap and waste. This digital-to-physical materialization process enables precise geometric articulation and adapts effectively to different load distribution scenarios.

The demonstrator was realized with the support of the Additive Manufacturing Center at TU Darmstadt and developed in a collaborative effort between the Digital Design Unit (DDU) of the Department of Architecture and the Institute of Printing Science and Technology (idd) of the Department of Mechanical Engineering at TU Darmstadt.



Plan



Bubbles choreography und assembly

FLOATING BUBBLES

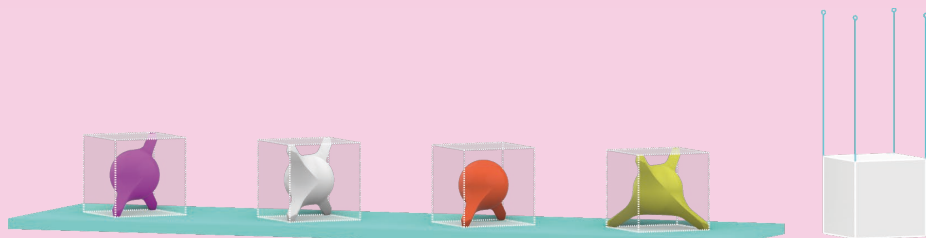
ON COLLABORATIVE REMOTE DESIGN FOR KINETIC TECTONICS
DIGITAL DESIGN STUDIO, SUMMER 2020
TECHNICAL UNIVERSITY OF DARMSTADT, DIGITAL DESIGN UNIT (DDU)
NASTASSIA SYSOYEVA, ANNA BRAUMANN, OLIVER TESSMANN, SAMIM MEHDIZADEH

This studio explores the potential of hollow architectural building blocks, which, unlike their massive counterparts that require heavy lifting equipment, can be easily moved, rotated, and interlocked. But what if these motions extended beyond the assembly phase and continued throughout their use? This investigation builds upon DDU's research on rotoformed, hollow, and lightweight concrete objects.

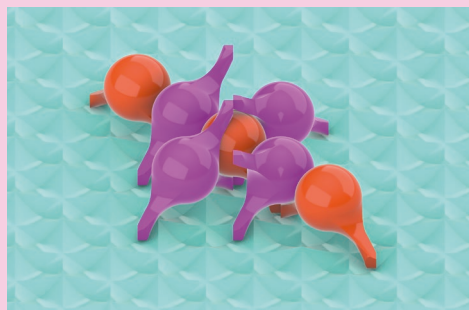
RotoForm creates objects that defy expectations—lighter than they appear and exhibiting counter-intuitive behavior due to a center of mass that deviates from their geometric centroid. The studio speculates on the functional and poetic possibilities of

these elements, envisioning them as dynamic, floating amusement structures that invite playful interaction.

Students engaged with computational design tools such as Rhino and Grasshopper, collaborated through parametric modeling with Speckle, and employed game engines to simulate physical behavior dynamically. Finally, Augmented Reality (AR) was used to create immersive experiences, allowing design proposals to be visualized within an urban context.



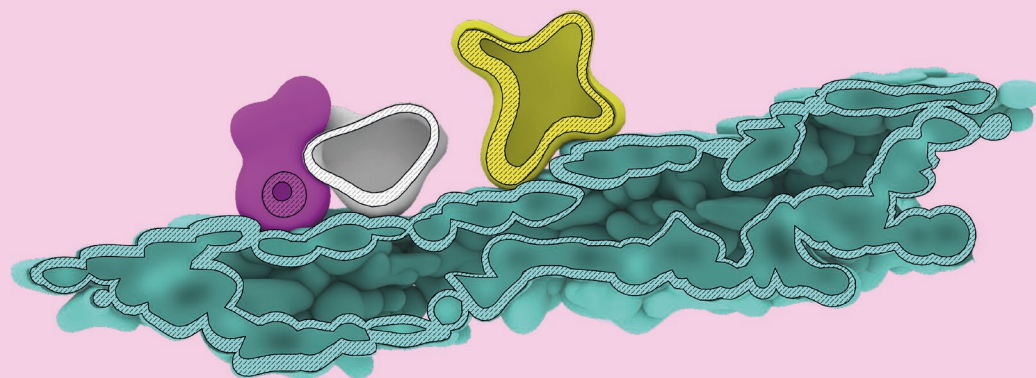
First generation of bubbles, inspired from Paul Schatz studies



Assembly possibilities



Floating Bubbles in AR app at the city in Frankfurt am Main



Ybble, Noseler and Star of the 2. generation, section

Floating Bubbles sind eine Familie skulpturaler architektonischer Formen, die durch einen gerundeten zentralen Körper und eine variable Anzahl von Auslegern gekennzeichnet sind. Zur Familie gehören Ubble (orange), Ybble (magenta), Xubble (weiß) und Sun (gelb), die sich jeweils durch ihre spezifische Form und Gliederkonfiguration unterscheiden. Das Team untersuchte ihr Schwimm- und Schwebeverhalten sowie den Einfluss des Massenmittelpunkts (Center of Mass, COM) auf ihre Bewegung, inspiriert von Paul Schatz' Studie „Die drei Symmetrien des Würfels, formkünstlerisch betrachtet“.

Eine zweite Generation dieser architektonischen Elemente entstand – die Blobbers, eine mutierte Weiterentwicklung der Bubbles. Zur Blobber-Familie gehören:

- Noselers (weiß): Benannt nach ihrer Ähnlichkeit mit Nasen eignen sich diese Formen besonders gut zum

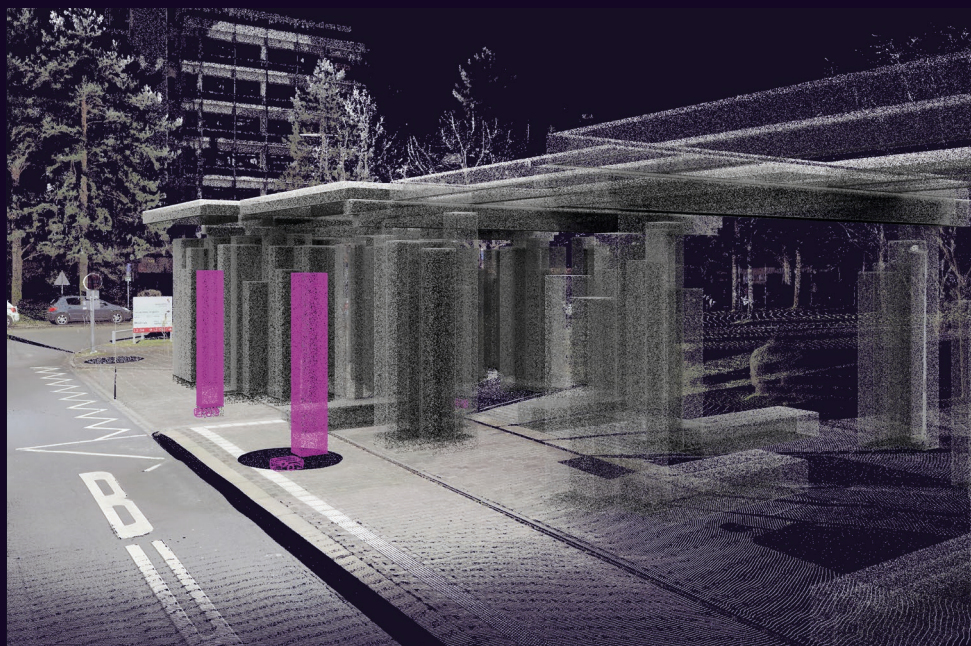
Schwimmen und Schweben.

- Ybblers (magenta): Gekennzeichnet durch drei Ausleger, die in einer Y-ähnlichen Form angeordnet sind.

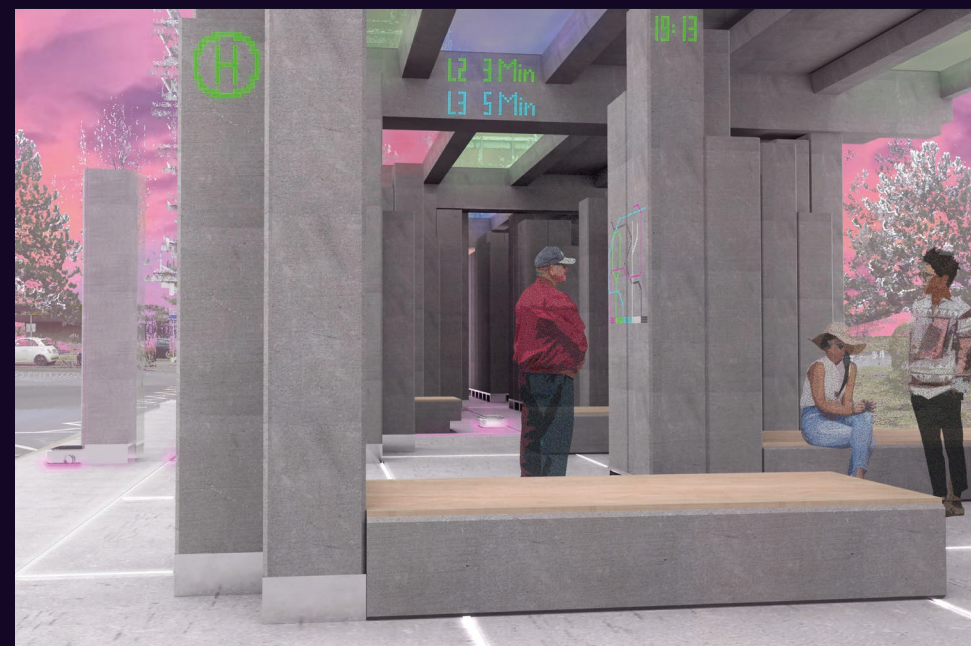
- Stars (gelb): Charakterisiert durch vier Arme, jedoch ohne zentralen Körper.

- Vobblers (orange): Bilden eine V-förmige Geometrie mit einem ausgeprägten zentralen Körper und zwei Armen.

Interessanterweise ist nicht immer eindeutig, ob es sich bei den Vorsprüngen um Arme oder um Körper handelt; durch ihre kompakten Ausleger eignen sich die Blobbers jedoch besonders gut dazu, sich gegenseitig zu umschließen. Diese intimen Interaktionen können zugleich als formschlüssige, trocken gefügte architektonische Verbindungen verstanden werden und verweisen auf neue Möglichkeiten des strukturellen Entwerfens.



Bus stop from a vehicles perspective



BUS STOP THROUGH THE LENSE OF A MACHINE

DESIGN FOR (RE)ASSEMBLY

DIGITAL DESIGN STUDIO, WINTER 2019

TECHNICAL UNIVERSITY OF DARMSTADT, DIGITAL DESIGN UNIT (DDU)

ANNA BRAUMANN, LUCIA MARTINOVIC, NASTASSIA SYSOYEVA

This studio explored resource-efficient circular economy strategies in architecture using digital and computational design tools. Disused buildings were envisioned as material banks: digitally scanned, converted into deconstruction models, and their structural elements harvested, subdivided, robotically cut, and reassembled with 3D-printed parts using Augmented Reality (AR). The workflow was prototyped in the DDU RoboLab through 3D scanning, photogrammetry, 3D printing, dry-joint connections, robotics, and AR. Students then applied these methods in the “Haltestelle der Zukunft” competition, launched by the city of Lörrach with IBA Basel 2020, which called for a future-oriented, cross-border bus stop concept.

Designs focused on reused concrete components, proposing the bus stop as a flexible repository for future buildings and illustrating innovative, sustainable material flows.

Our team 3D-scanned the TU Darmstadt architecture faculty building to explore reuse scenarios for its concrete structure. The project received an honorable mention in “FUTURE STOPPS – Halt 4.0.” The scanning was supported by Christian Hickel and the FG Geodätische Messsysteme und Sensorik under Prof. Andreas Eichhorn.



Scanned supports ready for demolition



1. HarvestMap

A building is listed on the website for the worldwide trade in used components.



Full process video:
<https://www.youtube.com/watch?v=6-bdJM5PEFo>

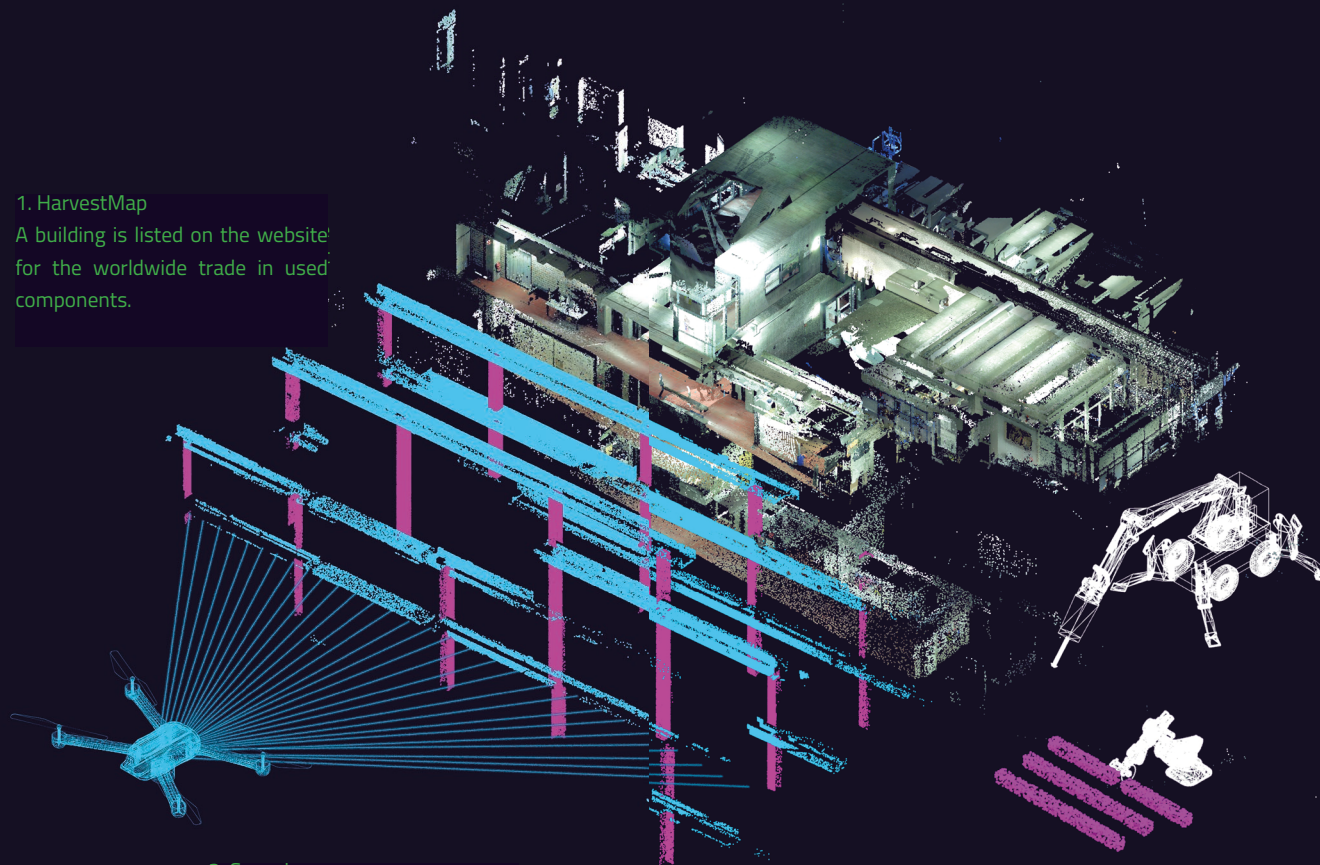


6. HarvestMap

Non-load-bearing supports will be available for sale and can be picked up from the stop.

7. Abbau

Supports ordered on HarvestMap can be removed and transported to a new construction site.



2. Scanning

A drone is scanning the building.

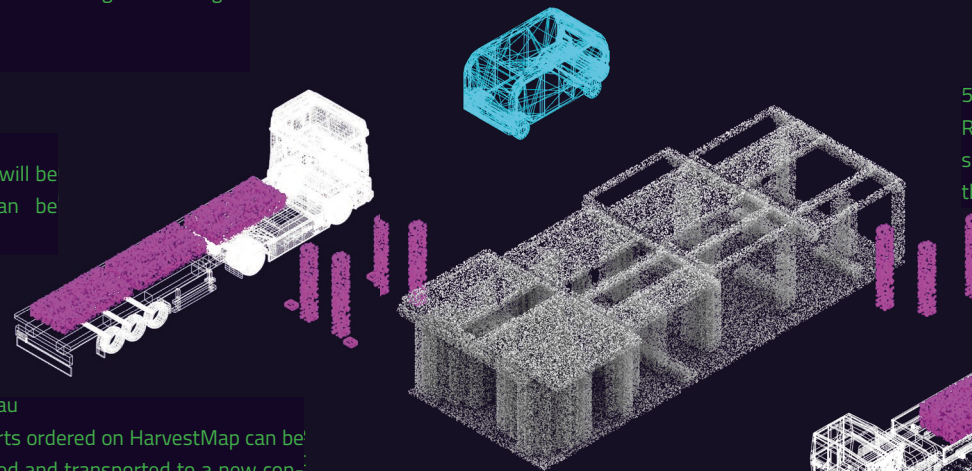
3. Demolition and cutting

The building is being dismantled and the parts that are still usable are being cut on site.



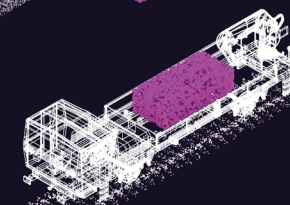
5. Assembly

Robots set up the stop, one side serves as storage for the supports.



4. Transport

Components arrive at the stop, e.g. by train.



Circular economy diagram"

